

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5				Indicative content temporary hard water contains hydrogencarbonate (HCO_3^-) ions whilst permanent hard water contains sulfate (SO_4^{2-})/other ions add soap solution to each sample shake each sample neither will produce a lather boil both samples the sample that now gives a lather on the addition of soap solution is temporary hard water / the sample that does not give a lather is permanent hard water method works due to hydrogencarbonate ions forming calcium carbonate/limescale on heating whilst sulfate/other ions are unaffected by heating 5-6 marks Good description of different composition and how to differentiate between the samples; clear understanding of the method <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Basic description of composition of either temporary or permanent hard water and how to differentiate between the samples; reference to 'furring' or precipitation of calcium carbonate <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Simple description of how to differentiate between the water samples <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>	5	1		6		6
				Question 5 total	5	1	0	6	0	6